

The Economic Aftershocks:

Analyzing the Impact of Major Earthquakes on National Recovery

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GitHub: https://github.com/betlm/DSA210_project

Motivation

Earthquakes are among the most destructive natural disasters on Earth, yet their economic consequences vary enormously across countries. A magnitude 7.0 earthquake can devastate a low-income country while causing comparatively manageable damage in a high-income one. This disparity has long been observed anecdotally, but systematic data driven evidence of the mechanisms behind it is less common in the academic literature.

My interest in this topic predates this project. Having previously explored earthquake-related data sources, including EM-DAT and the World Bank API, in earlier work, I was already familiar with the structure and richness of these datasets. This project gave me the opportunity to formalize those explorations into a rigorous data science analysis: combining multiple data sources, applying statistical hypothesis tests, and extending the analysis to machine learning.

The central question driving this project is whether earthquake intensity or a country's underlying economic resilience is the dominant factor in determining economic damage and recovery speed. The answer has direct policy implications: if wealth is the dominant factor, then disaster preparedness investment and institutional capacity-building matter more than any physical measure of seismic risk.

Data Sources

This project integrates three heterogeneous data sources, merged using Python into a unified dataset of 959 earthquake events across 110 countries from 1990 to 2026.

1. EM-DAT — Emergency Events Database

The primary dataset comes from EM-DAT (Centre for Research on the Epidemiology of Disasters), which records global disaster events including casualties, affected population, and economic damage. The dataset was filtered to earthquake events from 1990 onward. Key variables extracted include earthquake magnitude, total deaths, total affected population, nominal economic damage (thousands USD), and inflation-adjusted economic damage (thousands USD).

Property	Value
Provider	CRED — Centre for Research on the Epidemiology of Disasters
Events loaded	959 earthquakes
Countries covered	110
Year range	1990 – 2026
Key variables	magnitude, total_deaths, total_affected, total_damage_adj_kUSD

2. World Bank API

Three macroeconomic indicators were fetched for all countries from 1985 to 2030 via the World Bank REST API. A custom fetch function with retry logic (3 attempts, 5-second delay) and a 120-second timeout was implemented to handle network instability. Each indicator was pivoted into a country-by-year matrix for efficient event-level lookup.

Indicator Code	Metric	Rows Retrieved
NY.GDP.MKTP.KD.ZG	GDP Growth (annual %)	9,705
NY.GDP.PCAP.CD	GDP per Capita (current USD)	9,818
SL.UEM.TOTL.ZS	Unemployment Rate (%)	8,071

3. USGS Earthquake Catalog

The USGS global earthquake catalog (usgs_1900.csv) was used to enrich the EM-DAT events with focal depth (usgs_depth_km), a variable not recorded in EM-DAT. Each EM-DAT event was matched to the nearest USGS event in the same year using magnitude proximity (threshold: 0.5 magnitude units). Of 959 events, 751 (78%) were successfully matched.

Focal depth is an important predictor of surface damage: shallow earthquakes (< 70 km) produce much stronger ground shaking than deep ones at the same magnitude. This variable was included as a feature in the machine learning analysis.

Data Analysis

1. Data Merging and Preprocessing (Notebook 01)

The three data sources were merged in 01_data_merging.ipynb. For each earthquake event, three economic recovery metrics were computed from the World Bank GDP growth pivot table:

Metric	Definition
pre_event_avg_gdp	Mean GDP growth in the 3 years before the earthquake
gdp_drop	Event-year GDP growth minus the pre-event baseline
recovery_years	Years until GDP growth returned to or exceeded baseline (max 5 years; NaN if not recovered)

Income groups were assigned using World Bank 2024 thresholds applied to GDP per capita in the event year: Low (< \$1,085), Lower-Middle (\$1,085–\$4,255), Upper-Middle (\$4,255–\$13,205), High (> \$13,205). A significant bug was identified and corrected during development: a pandas index misalignment between the EM-DAT DataFrame (non-contiguous index from the raw Excel read) and the recovery metrics DataFrame (computed via apply on the filtered DataFrame) caused pd.concat to produce 1,918 rows, exactly double the correct 959, with all recovery metrics in the wrong rows and all country information missing from those rows. The fix was to call reset_index(drop=True) on both DataFrames before concatenation.

2. Exploratory Data Analysis (Notebook 02)

The EDA notebook explores three main themes: the distribution of earthquake magnitudes and economic damage, differences in economic vulnerability across income groups, and correlations between key variables.

Two derived variables were constructed for the analysis. log_damage is the log base 10 of inflation-adjusted damage in thousands USD, used because raw damage values span eight orders of magnitude. damage_ratio is total inflation-adjusted damage divided by GDP per capita, capturing the relative economic burden on citizens rather than absolute damage in dollars.

Variable	Count	Mean	Median	Max
magnitude	952	6.23	6.20	9.10
total_deaths	631	1,426	7	222,570
total_damage_adj_kUSD	396	\$3.28B	\$131M	\$293B
log_damage	396	5.08	5.12	8.47
damage_ratio	379	491	25.9	35,148
gdp_per_capita_event_year	889	\$7,343	\$3,093	\$82,139

The median damage ratio for Low-income countries (135.8) is approximately 8 times higher than for High-income countries (17.0), providing preliminary visual evidence for H2 before formal testing. The Spearman correlation matrix confirms that magnitude has a moderate positive correlation with log(damage), and GDP per capita is negatively correlated with damage ratio, wealthier countries absorb economic shocks more easily.

3. Hypothesis Testing (Notebook 03)

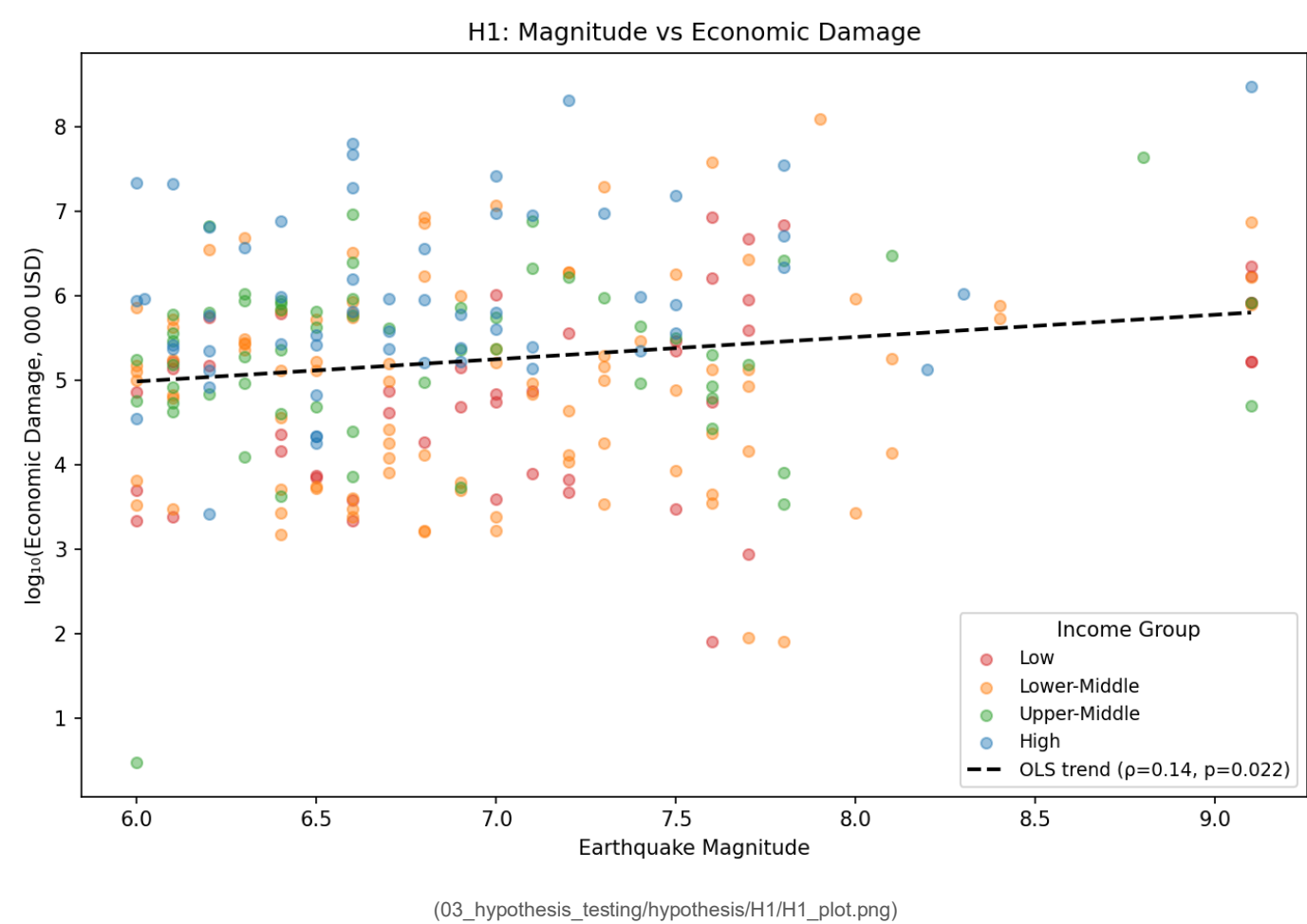
Two hypotheses were formally tested using non-parametric methods appropriate for the heavily skewed, outlier-rich distributions in earthquake economic data.

H1 — Magnitude vs Economic Damage

A Spearman rank correlation was computed between earthquake magnitude and log₁₀(inflation-adjusted damage) on the subset of M ≥ 6.0 events with recorded damage (n = 271). Spearman correlation was chosen over Pearson because the relationship may be non-linear and extreme outliers (e.g., the 2011 Tohoku earthquake) would distort Pearson estimates.

Metric	Value
n	271
Spearman rho	0.1640
p-value	0.006831
Decision	REJECT H0 — statistically significant positive correlation

The correlation is statistically significant but weak ($\rho = 0.16$), reflecting that magnitude alone explains only a fraction of economic damage variance. Factors such as population density, urban proximity, construction quality, and emergency response capacity play major roles that magnitude cannot capture. As shown in the scatter plot in 03_hypothesis_testing/hypothesis/H1/H1_plot.png, the OLS trend line confirms a positive but noisy relationship across all income groups.



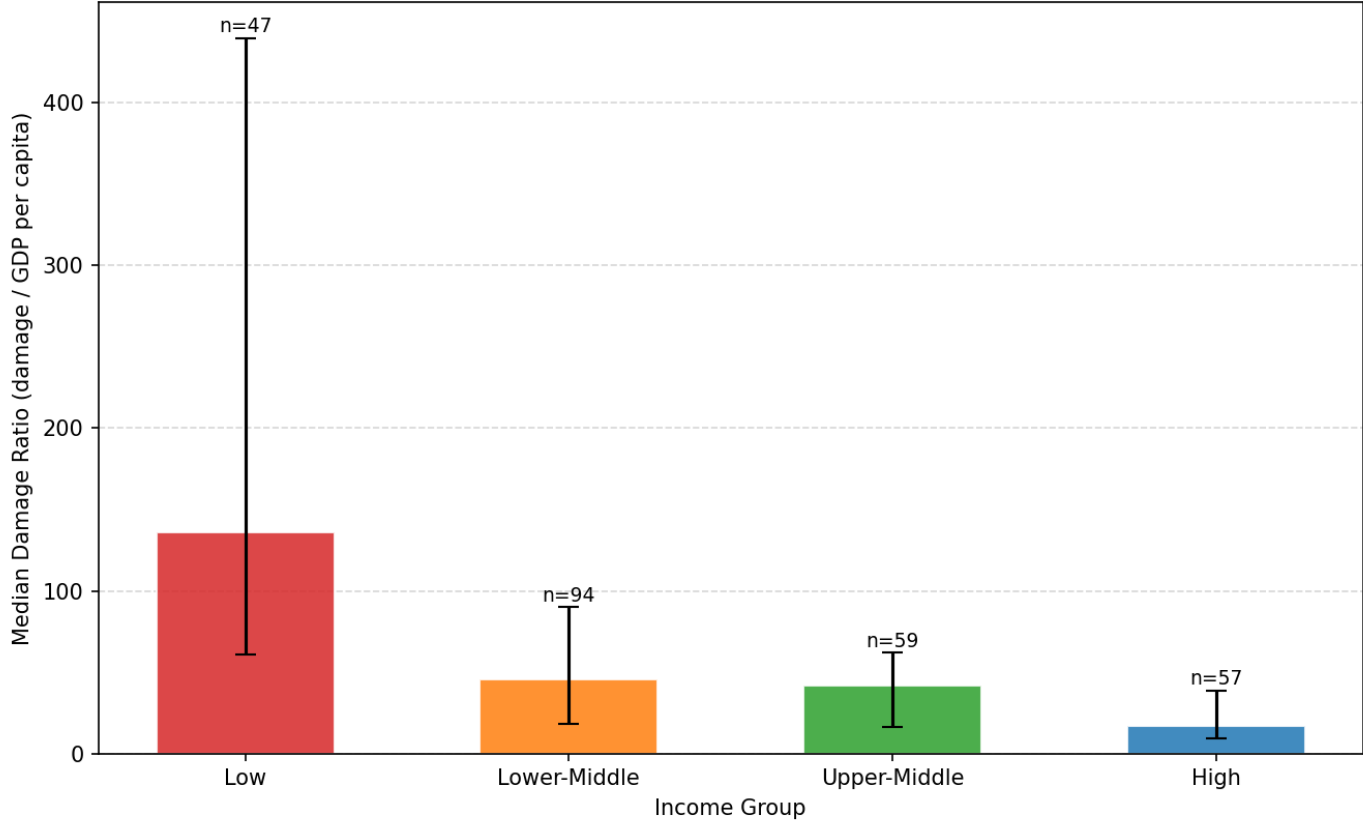
H2 — Income Level vs Economic Vulnerability

A Kruskal-Wallis test was applied to the damage ratio across four income groups on the $M \geq 6.0$ subset with computable damage ratios ($n = 257$). Kruskal-Wallis was chosen because damage ratio is severely right-skewed (mean = 491, median = 26) and does not follow a normal distribution, making ANOVA inappropriate.

Test	Statistic	p-value	Decision
Kruskal-Wallis	H = 9.750	0.0208	REJECT H0
Low vs Upper-Middle (Mann-Whitney, Bonferroni)	—	0.0308	Significant
Low vs High (Mann-Whitney, Bonferroni)	—	0.0715	Borderline
All other pairs	—	> 0.10	Not significant

The Kruskal-Wallis test confirms that income level significantly affects relative economic damage ($p = 0.021$). Post-hoc Bonferroni-corrected Mann-Whitney U tests identify the Low vs Upper-Middle pair as the only clearly significant pairwise difference. The Low vs High comparison, despite showing the largest median difference (135.8 vs 17.0 — an 8-fold gap), does not survive Bonferroni correction ($p = 0.072$), likely because the conservative correction combined with the small Low-income sample ($n = 47$) reduces statistical power. As shown in the bootstrap confidence interval chart in 03_hypothesis_testing/hypothesis/H2/H2_plot.png, the non-overlapping confidence intervals for Low and Upper-Middle income groups visually confirm the pairwise significance.

H2: Economic Vulnerability by Income Group
Median $\pm$ 95% Bootstrap CI



(03_hypothesis_testing/hypothesis/H2/H2_plot.png)

4. Machine Learning (Notebook 04)

The machine learning analysis tests H3: whether a nonlinear relationship exists between earthquake characteristics, preevent economic indicators, and recovery time that ensemble methods can capture better than linear ones.

Target Variable Reframing

An initial version of the analysis treated `recovery_years` as a continuous regression target. This version was discarded for two reasons. First, a target leakage issue was identified: `gdp_drop` and `recovery_years` are both derived from the same GDP growth time series in the data preparation step, meaning that predicting one from the other measures autocorrelation within the same series rather than genuine predictive ability. Second, `recovery_years` is bounded between 1 and 5 (right-censored at 5 years), making RMSE a misleading metric.

The target was reframed as a 3-class ordinal classification: Fast (`recovery_years` = 1), Medium (`recovery_years` = 2 or 3), and Slow (`recovery_years` >= 4). This is methodologically honest and aligns with the policy-relevant question of whether a country recovers quickly, moderately, or slowly.

Features

Feature	Type	Rationale
<code>magnitude</code>	numeric	Physical intensity of the earthquake
<code>usgs_depth_km</code>	numeric	Shallow quakes cause more surface damage
<code>gdp_per_capita_event_year</code>	numeric	Country wealth at event time
<code>unemployment_event_year</code>	numeric	Pre-existing economic fragility
<code>pre_event_avg_gdp</code>	numeric	Country's growth trend going into the event
<code>income_encoded</code>	ordinal	Coarser wealth signal (Low=0 to High=3)

Variables deliberately excluded: `gdp_drop` (target leakage), `total_deaths` / `total_affected` / `total_damage` (measured after the earthquake, not before), `gdp_event_year` (same time series as target).

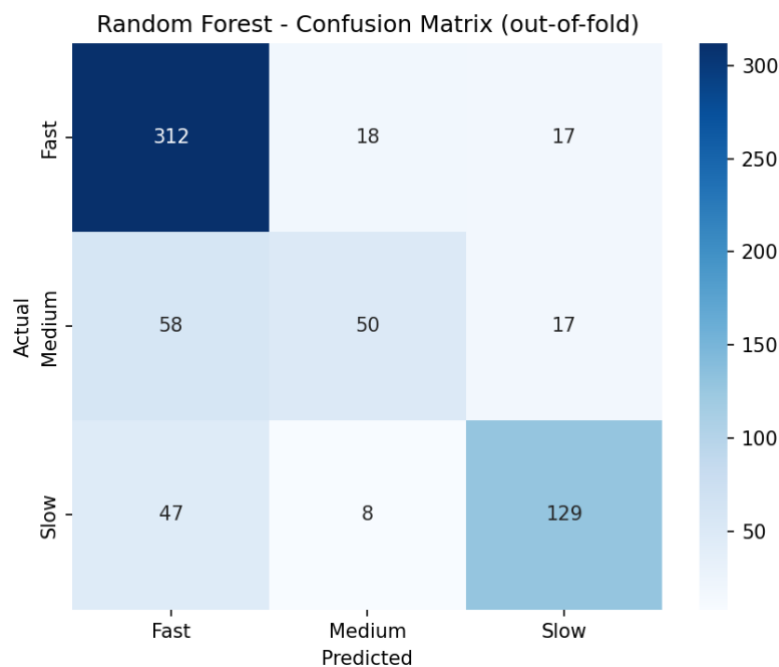
Evaluation and Results

Three models were evaluated using stratified 5-fold cross-validation on $n = 656$ events with complete pre-event features. A trivial majority class baseline was included as a sanity floor. A tuned Random Forest (GridSearchCV over `n_estimators`, `max_depth`, `min_samples_leaf`) was compared against Logistic Regression using a paired Wilcoxon signed-rank test on per-fold F1 scores.

Model	Accuracy	F1 Macro	vs Baseline
Trivial Baseline (always Fast)	0.529	0.231	—
Logistic Regression	0.668	0.477	+24.6 pp
Random Forest (tuned)	0.749	0.685	+45.4 pp

The Random Forest outperforms Logistic Regression by +8.1 percentage points in accuracy and +20.8 percentage points in macro-F1. A paired Wilcoxon test confirms this gap is statistically significant ($p = 0.031$), supporting H3. The superiority of Random Forest over Logistic Regression is consistent with H3's core claim: the relationship between pre-event economic indicators and recovery time is inherently nonlinear, income thresholds, growth trends, and unemployment interact in ways that decision trees capture through branching rules, while linear models cannot. Both models clear the trivial baseline by a

wide margin, confirming that the signal is real rather than an artifact of class imbalance. As shown in Figure in 04_machine_learning/outputs/confusion_matrix.png , the confusion matrix confirms the model's ability to distinguish Fast, Medium, and Slow recovery classes.



(04_machine_learning/outputs/confusion_matrix.png)

Feature Importance

Feature	Importance
pre_event_avg_gdp	35.9%
unemployment_event_year	20.2%
gdp_per_capita_event_year	18.9%
usgs_depth_km	11.6%
magnitude	10.0%
income_encoded	3.5%

The most striking finding is that physical earthquake characteristics (magnitude and depth together: 21.6%) are outweighed by pre-event economic conditions (pre_event_avg_gdp + unemployment + gdp_per_capita + income_encoded: 78.4%). Recovery time depends far more on the kind of economy the earthquake hits than on how strong the earthquake is.

Key Findings

All three hypotheses were supported by the data:

- H1 SUPPORTED: Higher earthquake magnitude is significantly associated with greater economic damage (Spearman rho = 0.164, p = 0.007). The effect is real but weak, indicating that magnitude is a necessary but not sufficient predictor of economic damage.
- H2 SUPPORTED: Income level significantly affects relative economic vulnerability (Kruskal-Wallis p = 0.021). Low-income countries show a median damage ratio 8 times higher than High-income

countries. Posthoc pairwise tests confirm the Low vs Upper-Middle gap is statistically significant; the Low vs High gap is practically large but falls just outside the Bonferroni corrected threshold due to sample size constraints.

- H3 SUPPORTED: A non-linear relationship exists between earthquake impact and economic recovery time that ensemble learning captures better than linear methods (Wilcoxon $p = 0.031$, macro-F1 improvement +20.8 pp). The Random Forest correctly classifies 74.9% of events into Fast, Medium, or Slow recovery, significantly above both the linear baseline and the trivial majority class predictor.

The overarching finding is that preevent economic conditions dominate earthquake outcomes. A country's growth trend, unemployment rate, and wealth level together account for nearly 80% of the feature importance in predicting recovery time, while the earthquake's physical intensity (magnitude and depth) accounts for only 21%. This is a striking reversal of intuition: it is the economy that determines resilience, not the earthquake that determines fate.

Limitations and Future Work

1. Data Limitations

- Missing damage data: Only 396 of 959 events (41%) have recorded economic damage values. This introduces selection bias catastrophic, well-publicized events are more likely to be reported, potentially inflating average damage estimates. Events from low-income countries are both more numerous and more likely to have incomplete data.
- Annual GDP resolution: Recovery metrics are computed from annual GDP growth data. Earthquakes occurring late in a fiscal year appear minimal in that year's data even if their damage is severe. Quarterly GDP data would provide more accurate recovery measurement.
- USGS depth matching: Focal depth was matched by magnitude proximity within the same year rather than by geographic coordinates, because latitude and longitude were not included in the EM-DAT export. This may introduce noise in the depth variable for countries that experienced multiple large earthquakes in the same year.
- is_tsunami flag: The tsunami indicator was set to 0 for all events because the EM-DAT export used did not include the Associated Types column. Future work should extract this field to capture the additional damage multiplier of tsunami generating earthquakes.

2. Methodological Limitations

- Bonferroni correction is conservative: The Low vs High pairwise comparison (uncorrected $p = 0.012$) did not survive Bonferroni correction ($p = 0.072$). The Benjamini-Hochberg false discovery rate correction would be less conservative and may be more appropriate for exploratory analysis.
- Target leakage in initial ML version: An initial regression model included gdp_drop as a feature when predicting recovery_years. Both variables are derived from the same GDP growth time series, constituting target leakage. This was identified and corrected by removing gdp_drop and reframing the target as 3-class classification. The correction reduced apparent R^2 substantially but produced honest, reproducible results.
- Confounding variables: Infrastructure quality, building codes, foreign aid flows, governance quality, and geographic factors (urban density, soil type) are not included in the model. These likely explain a substantial portion of the remaining unexplained variance.
- Right-censoring of recovery_years: Events where GDP growth did not return to baseline within 5 years are coded as NaN and excluded from the ML analysis. This excludes the most economically devastated cases and may underestimate recovery difficulty for the worst-affected countries.

3. Future Work

- Add infrastructure and governance indicators (e.g., World Bank Governance Indicators, Global Infrastructure Index) as features.
- Use quarterly GDP data where available to improve recovery metric precision.
- Apply Benjamini-Hochberg correction as an alternative to Bonferroni for hypothesis testing.
- Extend the ML analysis with time-series models that capture recovery trajectories rather than a single recovery year classification.
- Incorporate insurance loss data (e.g., Swiss Re Sigma) to supplement EM-DAT's incomplete damage reporting.

Conclusion

This project demonstrates that earthquake economic outcomes are determined far more by a country's pre-existing economic conditions than by the physical intensity of the seismic event. While magnitude is a statistically significant predictor of damage (H1), and income level significantly shapes relative vulnerability (H2), the machine learning analysis reveals that pre-event GDP growth trend, unemployment, and wealth level together account for nearly 80% of the predictive signal for recovery time (H3).

These findings have direct policy implications. Investing in economic resilience, stable growth, low unemployment, strong institutions, may be as important as investing in physical earthquake preparedness. A country with a strong economy recovers from a large earthquake faster than a poor country recovers from a small one. Data driven evidence of this pattern, at the scale of 959 events across 110 countries over 35 years, provides a rigorous empirical foundation for that claim.

AI Usage Disclosure

AI assistance (Claude by Anthropic) was used throughout this project. In accordance with the DSA 210 Academic Integrity policy, all AI-assisted contributions are disclosed below:

- Bug fixing: AI identified the root cause of a pandas index misalignment bug that caused 959 phantom rows in the merged dataset and all recovery metrics to be placed in wrong rows.
- Target leakage identification: AI identified that `gdp_drop` and `recovery_years` were derived from the same GDP time series, constituting target leakage in the initial ML version.
- Methodology: AI clarified why Spearman correlation is preferred over Pearson for skewed damage data, why Kruskal-Wallis is appropriate instead of ANOVA for non-normal distributions, and why RMSE is a misleading metric for bounded integer targets.
- ML reframing: AI proposed the 3-class classification reframing, the trivial majority class baseline, and the paired Wilcoxon significance test for H3.
- Code assistance: Syntax help for World Bank API fetch logic, `scipy.stats` functions, sklearn pipelines, and bootstrap confidence interval calculation.
- Documentation: Writing assistance for markdown cells in notebooks, README, and this final report.

All analytical decisions, hypothesis design, choice of variables, test selection, interpretation of results, and conclusions, were made by the author. All AI-assisted code and text was reviewed and approved before inclusion.